

COURSE TITLE : FABRIC FORMATION III
COURSE CODE : 5061
COURSE CATEGORY : A
PERIODS/WEEK : 5
PERIODS/SEMESTER : 65
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Multiple Box Looms, Terry Looms and Automatic Looms	15
2	Advanced Fabric Manufacture	16
3	Jet Weaving	17
4	Calculations	17
TOTAL		65

COURSE OUTCOME

Module.	GO	Student will be able to
1	1	Understand the features of Multiple Box Looms
	2	Know clear idea of terry motion in power loom
	3	Understand the features of Automatic Looms
2	1	Know the features of shuttle less looms and Sulzer shuttle less weaving.
	2	Understand the features of Rapier looms
3	1	Know the features of air jet looms
	2	Understand various the features of water jet looms
4	1	Understand the speed , production & efficiency calculations of various power looms

SPECIFIC OUTCOME

MODULE I MULTIPLE BOX LOMS,TERY LOOMS AND AUTOMATIC LOOMS

1.1.0 Understand the features of Multiple Box Looms

- 1.1.1 State the use of multiple box looms.
- 1.1.2 List the the various classification of Multiple box looms
- 1.1.3 Distinguish between Drop box and Circular boxes.
- 1.1.4 Differentiate Skip and Non-skip Multiple box looms.

- 1.1.5 Discuss the Pick at will looms
- 1.1.6 Explain the working of Eccle's Drop box motion and card saving mechanism.
- 1.1.7 Explain the Arrangement of pattern cards for various picking pattern in Eccle's drop box motion
- 1.1.4 Describe the Working of non-skip circular box motion.
- 1.2.0 Know clear idea of terry motion in power loom**
 - 1.2.1 State the Objectives of Terry motion.
 - 1.2.2 List the classification of Terry motion
 - 1.2.3 Explain the working of Dugdale's Terry motion
 - 1.2.4 Describe the working of Holden's Terry motion.
- 1.3.0 Understand the features of Automatic Looms**
 - 1.3.1 List the classification of automatic looms.
 - 1.3.2 State the Features of automatic looms.
 - 1.3.3 Compare automatic and non- automatic looms
 - 1.3.4 Describe the working of Midget weft feeler mechanism
 - 1.3.5 Explain the working of automatic weft replenishment motion with reference to cop changing automatic looms
 - 1.3.6 Describe the working of Shuttle protector and its use
 - 1.3.7 State the use of thread cutters
 - 1.3.8 List the Features of Shuttle changing looms

MODULE II ADVANCED FABRIC MANUFACTURE

- 2.1.0 Know the features of shuttle less looms and Sulzer shuttle less weaving**
 - 2.1.1 State the Features of shuttle less looms
 - 2.1.2 Compare conventional method of weaving and shuttle less weaving
 - 2.1.3 Classify shuttle less looms
 - 2.1.4 Describe the Basic requirement for shuttle less weaving
 - 2.1.5 List the Advantages of Shuttle less weaving.
 - 2.1.6 State the principles of Sulzer shuttle less weaving
 - 2.1.7 Illustrate the various stages of weft insertion in Sulzer shuttle less –weaving
 - 2.1.8 Explain the working of torsion bar picking.
 - 2.1.9 Describe Sulzer Shuttle less weaving machine with 4 colour weft insertion
 - 2.1.10 List the quality of fabrics produced on Sulzer looms.
 - 2.1.11 Discuss the working of cam beat –up mechanism
- 2.2.0 Understand the features of Rapier looms**
 - 2.2.1 State the features of Rapier shuttle less weaving.
 - 2.2.2 List the Various methods of weft insertion in Rapier looms
 - 2.2.3 Explain various stages of weft insertion in a flexible Rapier loom.
 - 2.2.4 Describe the Rack and pinion type Picking mechanism in a flexible Rapier loom
 - 2.2.5 Describe the selvedge like leno selvedge, tuck- in selvedge, looped selvedge, fused selvedge, glued selvedge
 - 2.2.6 State Merits and Demerits of Rapier weaving.

MODULE III JET WEAVING

3.1.0 Know the features of air jet looms

- 3.1.1 Discuss the features of Air Jet Looms.
- 3.1.2 Explain the Passage of warp yarn on a Maxbo-air Jet loom
- 3.1.3 State the Function of weft insertion elements in air jet loom.
- 3.1.4 Describe the the various stages of weft insertion in Maxbo air jet loom

3.2.0 Understand various the features of water jet looms

- 3.2.1 State the Features of water jet loom
- 3.2.2 Illustrate 4 stages of weft insertion in water jet looms.
- 3.2.3 Discuss the working of weft supply system on water Jet looms.
- 3.2.4 Explain the water pump, Throttle value and Water Jet nozzle
- 3.2.5 List the Merits & Demerits of water jet looms
- 3.2.6 Draw a table which shows the comparison of popular shuttle less looms

MODULE IV CALCULATIONS

4.1.0 Understand the speed , production & efficiency calculation of various power looms

- 4.1.1 Determine the speed of various types of looms
- 4.1.2 Estimate the production of different types of looms.
- 4.1.2 Estimate the efficiency of various types of looms
- 4.1.3 Find out the time required to exhaust a weaver's beam.
- 4.1.4 Determine the time required for the exhaustion of weft packages
- 4.1.5 Find out the number of pirns required for a loom to run a particular time
- 4.1.6 Find out the quantity of pirn/cone/cheese etc. required to produce a particular quantity of fabric
- 4.1.7 Calculate the number of pirn winders required to feed given number of looms
- 4.1.8 Determine the number of warping machine required to run a particular number of looms .
- 4.1.9 Design charts showing the loom speeds for various width of plain and automatic looms .
- 4.1.10 Find out the quantity of warp & weft yarn required to weave fabrics having various widths
- 4.1.11 Estimate the total weight of yarn required to produce a particular type of fabric
- 4.1.12 Determine the cost of different types and quality of fabrics.

CONTENT DETAILS

MODULE – I MULTIPLE BOX LOOMS, TERRY MOTIONS AND AUTOMATIC LOOMS.

Introduction to weft patterning-Type of box motions-Use of multiple box looms-Classification of multiple boxes-Drop boxes Vs Circular boxes-Skip Vs Nonskip-Pick at will/Pick and Pick. Working of Eccle's Drop box motion-Study of card saving mechanism-Arrangement of pattern cards for various picking patterns-Study of various positions of stud and pin when different boxes are in line with the slay race-Timing and setting. Working of Non- Skip type circular box motion-Object and classification of Terry motions-Working of Dugdale's Terry motion-Working of Holden's Terry motion-Features of automatic looms-Classification of automatic looms-Advantages of automatic looms-Compare automatic and non-automatic looms. Working of midget weft feeler mechanism-Timing and setting-Features of two prong electrical feeler and photo electric feeler. Working of automatic weft replenishment motion with respect to cop changing looms - Timing and settings. Shuttle protector –functions and uses. Different types of Thread cutters and their uses-Features of shuttle changing looms. Compare shuttle changers and cop changers.

MODULE –II ADVANCED FABRIC MANUFACTURE

Introduction on various methods of fabric production-Compare the advanced method of fabric production with conventional types of looms-Mention the types of shuttle less looms-Mention the basic requirement of shuttle less weaving-Mention the advantages of shuttle less weaving machines. Mention the principles and explain the various stages of weft insertion in Sulzer shuttle less weaving machine with 4 colour weft insertion -Explain the Torsion bar picking mechanism-Explain the working of cam beat –up. Mention the quality of Sulzer woven cloth-Give brief note on warp and weft preparation for shuttle less weaving-Give introduction to Rapier shuttle less weaving-Variations of Rapier shuttle less looms-Mention the merits, demerits and limitations of Rapier looms-Explain the various stages of weft insertion in rapier looms-Explain the picking mechanism in a flexible rapier loom (Rack and pinion type)-Features of different types of selvages used in shuttle less weaving.

MODULE –III JET WEAVING

Explain the features of Air Jet looms-Explain the passage of warp yarn on Maxbo Air Jet loom-Draw the diagram of Air Jet nozzle for weft insertion in air jet looms-Explain the functions weft insertion elements used in air jet looms-Explain the various stages of weft insertion on Maxbo air jet looms-Explain the basic features of water Jet looms-Explain the 4 stages of weft insertion in water jet loom-Explain the working of weft supply system on water jet looms-Draw the diagram of the pump, throttle valve and water jet nozzle-Mention the merits, demerits and limitations of water jet looms. Give the comparison charts of popular shuttleless looms.

MODULE–IV CALCULATIONS

Calculate the production and efficiency of power looms-Evaluate the times required for the exhaustion of a weavers beam-Evaluate the time required for the exhaustion of a pirn, cone etc-Estimate the

number of pirns required for the loom in a given time-Estimate the number of pirns / cones required to produce a particular quantity of fabric-Compute the number of pirn winders required to feed a given number of looms-Estimate the number of warping machines required to feed the given number of looms-Prepare a chart showing the standard loom speeds for various widths of plain and automatic looms-Find out the quantity of warp and weft required to weave a particular fabric. Find out the cost of different types of fabrics.

REFERENCE BOOKS

1. Thomas W. Fox The Mechanism of Weaving Universal Publishing corp.,Mumbai -2
2. N.N.Banerjee Weaving Mechanism Vol - I Smt Tandra banerjee & SmtApurba banerjee ,
West Bengal
3. NCUTE Woven Fabric Production II NCUTE New Delhi
4. R.Sengupta Weaving Calculations D.B.Taraporevala sons & co Ltd., Mumbai
5. Prof. J.C.Chakravarthy Mechanism of Weaving Vol- I&II Prof. J.C.Chakravarthy West Bengal